

SPECIAL ISSUE PAPER

Mobile medical app development with a focus on traceability

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Abstract

Today, the growth of medical devices and mobile medical applications is increasing enormously, thanks to the efficiency and enhancement of new technology. When it comes to mobile medical apps, developers need to understand what is required when a mobile application fulfils the definition of a medical device. Such applications have to be developed in compliance with medical device regulations. This can be a challenge for mobile medical application developers, as medical device software is normally developed in a manner that will also ensure the production of regulatory documentation that is essential to market such devices. In this paper, we identify the need for a mobile medical application development framework, the key criteria for such a framework, and describe how the results were collected through performing a Medical Device Software Development workshop. Furthermore, we describe how MDevSPICE together with an agile software development approach can be tailored to support a mobile medical applications development framework. We detail one of the key criteria for mobile medical application development framework—traceability.

KEYWORDS

medical device, medical device software, mobile application, mobile medical app, TAIF model, traceability

1 | INTRODUCTION

Today, medical device (MD) software systems, including mobile medical apps (MMAs), play an important role in our lives. Medical device software, according to the Food and Drugs Administration (FDA) definition, is a “software system that has been developed for the purpose of being incorporated into the medical device being developed or that is intended for use as a medical device in its own right”.¹ However, an MMA is “a mobile app that meets the definition of device in section 201(h) of the Federal Food, Drug, and Cosmetic Act (FD&C Act) 4; and either is intended to be used as an accessory to a regulated medical device or to transform a mobile platform into a regulated medical device.”² “If a mobile app is intended for use in performing a medical device (MD) function (ie, for diagnosis of disease or other conditions, or the cure, mitigation, treatment, or prevention of disease) it is a medical device.”²

Given the safety-critical nature of MDs, standards are in place to ensure the safety of these devices. So as to market an MD, it is

mandatory to satisfy the regulatory requirements within that particular region. Previously, the authors developed MDevSPICE (henceforth referred to as MDevSPICE), which contains all the MD software development best practices³ and most of the standards in 1 place. This will be discussed in more detail further in the paper as by tailoring MDevSPICE for MMAs—we envisage developers using it as their main development approach for MMAs.

1.1 | Problem statement

The challenge that MMA software development companies face when they want to market an app is the adherence to the large number of regulatory requirements specified in various international standards. To help these companies better prepare for demanding and costly regulatory audits alongside the development process, we propose to develop an MMA development framework. This framework would ease, assist the development process of an app, and facilitate the collection of all regulatory requirements by guiding developers from the initial requirements gathering right through to the finished product.

Within this paper, we discuss how MMA development with an agile approach may be introduced into MDevSPICE and finally create

an MMA development framework that would comply with all the regulations and allow developers to successfully market their application.⁴

Section 2 looks at the different MD regulations and classifications for MD and MMA. In Section 3, we describe MDevSPICE. In Section 4, we describe the methodology adopted to develop the MMA development framework and we describe the workshop that was performed to discuss the different software development approaches for MD software. In Section 5, we focus on one of the MMA development framework criteria that emerged from the workshop—traceability. We look at its benefits and barriers and describe the traceability assessment and implementation framework (TAIF) that was developed by some of the authors and validated to assist MD companies. The paper concludes with details of our future work in Section 6.

2 | MEDICAL DEVICE REGULATIONS

The largest global bodies responsible for issuing and managing MD regulations are the FDA in the United States and the European Commission in the EU.

- The FDA issues the regulation through a series of official channels, including the Code of Federal Regulation title 21, chapter I, subchapter H, part 820.⁵
- In the EU, the corresponding regulation is outlined in a number of directives: the general medical device directive (MDD) 93/42/EEC⁶; the active implantable MDD 90/385/EEC⁷; and the in vitro diagnostic MDD 98/79/EC⁸—all 3 of which have been amended by 2007/47/EC.⁹

2.1 | Medical device software regulations

The safety classifications for MDs depend upon the particular geographical region; the 2 that we refer to are the United States and Europe. The MD safety classification relates to the clinical safety of the device.

Within the safety-critical domains, it is important to have a quality management system (QMS) in place. For example, the QMS required to meet the regulatory requirements of the MD domain in the EU is ISO 13485.¹⁰ ISO 13485 is a specialization¹¹ of ISO 9001 for the MD domain. Quality management system requirements need to be adhered to in order to market an MD anywhere in the world. As ISO 13485 does not target software requirements, it can be used in conjunction¹² with IEC 62304 to support the safe design and maintenance of MD software. IEC 62304 is a critical standard for MD software developers, as it is the only standard that provides recommendations for MD software implementations based on the worst-case scenario of software failure causing hazards. For general MD risk management, IEC 62304 is used in conjunction with ISO 14971¹³ and IEC 80002-1.¹² IEC 80002-1 provides guidance on the application of ISO 14971 for software development.

IEC 62304 considers an MD to include software as opposed to being a complete MD system in its own right. The system or product level requirements are not included within IEC 62304 but instead

within the MD product standard IEC 60601-1.¹² Because of the increasing importance of usability of devices within the MD industry, organizations should also adhere to the MD usability engineering process requirements outlined in IEC 62366.¹⁴

Stand-alone software systems (such as MMAs) may be defined as a MD in their own right, due to an amendment to the MDDs.⁹ However, until recently, international MD standards did not address stand-alone software as an MD. To address this, IEC FDIS 82304-1¹⁴ will soon be published. IEC FDIS 82304-1 applies to the safety of health software that is designed to operate on general purpose IT platforms intended to be placed on the market without dedicated hardware, eg, tablet and phone applications. Therefore, IEC FDIS 82304-1 will be applicable to MMAs.

If the company is planning to market a MD in the United States, they need to register their product with the FDA. The FDA has issued an overview of their guidance documents for developers of MD software.¹⁵ There are 4 guidance documents: the FDA Guidance on Premarket Submissions¹⁶; the FDA Guidance on Off-The-Shelf Software Use in MDs¹⁷; the FDA General Principles of Software Validation¹⁸; and Guidance on MMAs¹ to provide clarity for MMA developers in terms of what types of applications will be regulated and how.

Whenever companies developing MD software wish to market a device, they must adhere to a large number of regulatory requirements and international standards. In the case of MMAs meeting the definition of an MD, they have to comply with the same regulations as an MD. To reduce the demanding and costly overhead associated with preparing for regulatory audits, we developed MDevSPICE.¹⁹ MDevSPICE integrates the software requirements from all the above mentioned international MD standards and guidance documents with generic software development best practices into a single reference source. This reference source may then be used to assess the capability of an organization's MD software processes.

2.1.1 | Medical device software classifications

IEC 62304:2006¹² describes MD software life cycle processes and outlines software safety classification, based on the worst possible consequence in the case of a software failure. In the case of failure of software that is of safety Class A no injury or damage to health of a patient can occur. When software of safety class B fails, injury may occur but it is not serious or life threatening. Class C is of the highest risk and in the case of failure of such software death or serious injury can happen. Depending on the functionality of software within the MD, the software safety classification may vary from the overall MD safety class. When software is of critical functionality of the MD, it will carry the same classification as the device, ie, Class C software in Class III device.

2.2 | Mobile medical app types

Both the United States and Europe have released guidelines “to support developers and manufacturers ensure public safety.”²⁰

In September 2013, the FDA in the United States issued a final guidance on MMAs.¹ The FDA categorized MA into 3 categories depending on their risk to a human in case of its failure:

1. Those apps meeting the definition of a MD and “either is intended to be used as an accessory to a regulated MD; or to transform a mobile platform into a regulated MD,”¹ therefore “are the focus of FDA’s regulatory oversight”¹ These mobile apps pose high risk to a human in case of its failure—ie, injury or death. An example of an app for this category would be one that regulates infusion pump settings;
2. Those apps meeting the definition of a MD, however, through posing a low risk to a human in case of its failure, the FDA intends to exercise enforcement discretion.¹ For example, an app that helps an asthmatic to track their inhaler usage;
3. The third category includes all apps that do not meet the definition of a MD; therefore, the FDA does not regulate them. For example, an app that educates patient or tracks their fitness progression. To sum it up, if a mobile app cannot in any way harm a human, the FDA will not subject it to regulations.

In Europe, in January 2012, MEDDEV 2.1/6 was published to provide guidelines on stand-alone software development for MMAs.²¹ So as to market an app as an MMA, a CE Mark is required indicating that the requirements of either the MDD or the in vitro diagnostic medical device (IVDD) have been fulfilled.

3 | WHAT IS MDEVSPICE?

MDevSPICE was designed to assist manufacturers to develop MD software through following a traditional waterfall model or V-model life cycle model. Such development life cycles are very rigid, prescribed, and sequenced, which makes them both unsuitable and slow for MMAs development.

As IEC 62304 describes only the software life cycle processes, additional processes are required when the software developed forms the complete MD. In such cases, systems development processes need also to be considered. These systems development processes were derived²² from ISO/IEC 12207:2008. Quality Management and Risk Management related requirements from ISO 13485 and ISO 14971 were also added to MDevSPICE as both ISO 13485 and ISO 14971 are de facto standards for MD software development organizations. MDevSPICE consists of 23 processes, see Table 1. Ten are system life cycle processes, 8 are software life cycle processes, and 5 support both the system and life cycle processes.

MDevSPICE was validated through both international expert review and industrial trials. The foundation of MDevSPICE is IEC TR 80002-3, the process reference model (PRM) for IEC 62304.²³ Upon

successful completion of international expert reviews, MDevSPICE was validated through pilot assessments in 10 MD software development organizations over 2 years.

MDevSPICE provides full life cycle coverage in a plan-driven manner describing the requirements of all associated MD standards in 1 place, integrating software engineering best practices so that safer MD software may be developed.

However, the emergence of MMAs presents a new challenge to the development of MD software is with.

4 | APPROACH USED TO UNDERSTAND THE NEEDS FOR MMA DEVELOPMENT FRAMEWORK

The research questions we were exploring are as follows:

- What are the challenges associated with development of MMA software?
- What are the criteria/features developers would like to see within MMA development software framework?
- What are the existing software development approaches for developing MMA software?

To answer these questions, we performed a workshop that included the experts from the MD community. The Oxford dictionary defines a workshop as “a meeting at which a group of people engage in intensive discussion and activity on a particular subject or project.”²⁴ An action research methodology consists of 5 phases: diagnosing, action planning, action taking, evaluating, and specifying learning²⁵ and will be described in more detail in the following subsections. In the development of the MMA framework, the research methodology adopted is action research and the workshop formed a key part of the diagnosing and action planning stages.

4.1 | Mobile medical app development framework workshop

To be able to run a truly scientific workshop, we approached 37 organizations that develop either MD or MMA software. The Regulated Software Research Centre maintains a repository of contact details of MD software experts within Ireland, and this was used as the basis for identifying suitable organizations. The selection process then consisted of inviting representatives (from this repository) from

TABLE 1 MDevSPICE processes

Medical Device System Lifecycle Processes	Medical Device Software Lifecycle Processes	Medical Device Support Processes
Project management	Software development planning	Configuration management
Risk management	Software requirements analysis	Software release
Stakeholder requirements definition	Software architecture design	Software problem resolution
Project assessment and control	Software detailed design	Software change request management
System requirements analysis	Software risk management	Software maintenance
System architecture design	Software unit implementation and verification	
System integration	Software integration and integration testing	
System qualification testing	Software system testing	
Software installation		
Software acceptance support		

multinational and small to medium enterprises (SMEs) MD software communities within Ireland.

Fifteen (41%) of the invited organizations sent at least 1 software development representative, resulting in a total of 23 representatives. We ran 2 workshops in 2 different locations, Dundalk (North of Ireland) and Limerick (South of Ireland) to cover the geographical spread, see Table 2. These workshops were the diagnosing stage of our research methodology.

A structured invitation template explaining the objective of the workshop and location was developed along with a separate word document detailing the agenda that explained the workshop structure. These details were sent to chosen experts approximately a month prior to the first workshop. After sending invitations, a reminder email was sent 1 week prior to the first workshop, followed by an email to the confirmed representatives detailing location (including a map and the conference room number). Both workshops were run in academic institutes—Dundalk Institute of Technology and the University of Limerick; we discussed the changing face of MD software and the associated standards, and regulations that organizations need to be aware of today. We also looked at the current MMA software development approaches and discussed the challenges when developing MMAs. Then, we had a session discussing the awareness levels of agile for MMA software development, and we concluded the workshop with suggestions with what should be included in an MMA development framework.

4.2 | Mobile medical application development challenges

One of the discussion topics within the workshop was, What are the challenges associated with developing MMA software? Participants were divided into groups of 4 to 5 people from different companies and were allocated 20 minutes for discussion. At the end of this discussion, group leaders were asked to preset the list of MMA dev challenges on a flipchart. Once all challenges were reported back, an overall discussion took place to enable us to arrive at the final agreed list of challenges, which are shown in Table 3.

TABLE 2 Workshop attendance Figure 1

Location	Attendees	Organizations
Dundalk	13	9
Limerick	10	6
Total	23	15

TABLE 3 MMA development challenges in A to Z order

• Adoption/uptake • Architecture • Automated testing tools • Best available development toolset • Connectivity, security and the validation of them • Data flow • Dependency on OS, hardware, and network • Development lifecycle and app distribution • Hardware/platform, eg, phone/tablet, iOS/android • Knowing if app is a component or a complete MD	• Overhead of adopting required standards and compliance • Platform selection and validation • Providing essential updates when offline • Release and version management • Synchronisation between different devices for apps • Time to market • Understanding what SOUP is • Usability and testing on different versions of app hardware and platforms
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Abbreviations: MD, medical device; MMA, mobile medical app; SOUP, software of unknown provenance.

4.3 | Mobile medical app development framework features

Our second workshop discussion related to the following question: What are the criteria/features developers would like to see within an MMA development software framework? This discussion was run in the same manner as outlined in Section 4.2.

We discovered that there is a necessity to develop regulatory compliant applications in a shorter time frame than traditional MD software. Figure 1 represents the features that experts considered should be present into an MMA development framework. As illustrated in Figure 1, the most highly desired features were regulatory requirements, the need for documentation to satisfy these requirements through objective evidence and traceability.

To be able to market an MMA, we find that it is essential to achieve regulatory compliance. This involves providing fully traceable documentation that forms the objective evidence required by the auditors. Therefore, it is understandable that these features: documentation, adaption to regulatory requirements, knowledge of regulations, and traceability were the most sought after from the MD software experts that participated in the workshop. However, we consider “time to market” to also be important and a key distinguishing factor between the development of traditional MD software and the development of MMAs.

The workshop outlined that the mentioned features could be achieved through a hybrid development approach when agile practices were integrated into a plan driven software development approach such as the V-model.

4.4 | Approaches

The action planning stage of the action research methodology involved a third workshop discussion related to the following question: What are the existing software development approaches for developing MMA software? This discussion was conducted as outlined in Section 4.2.

Generally, in the MD software industry, the V-model is adopted.²⁶ However, due to successes reported within other domains from implementing agile practices, an international working group was set up to investigate if agile practices could fulfil the requirements of IEC 62304. The Association for the Advancement for Medical Instrumentation working group found that “agile practices can be successfully adopted to develop regulatory compliant software”²⁷ and produced AAMI:TIR 45:2012,²⁸ which is an internationally accepted technical

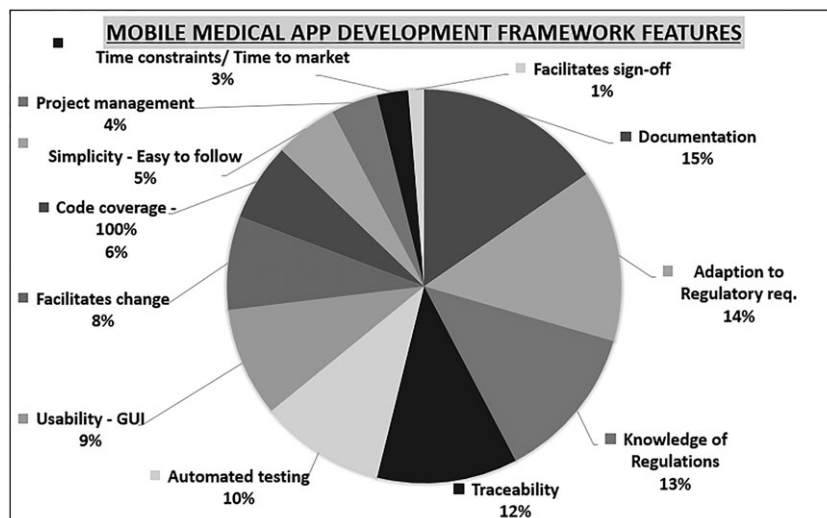


FIGURE 1 Mobile medical app development framework features

report called “Guidance on the use of Agile practices in the development of medical device software.” AAMI:TIR 45 provides general guidelines on how agile methods may be used to satisfy IEC 62304. To obtain a suitable development life cycle for MMAs, we have compared various potential MD software development approaches before deciding to develop an MMA framework where agile practices are combined with MDevSPICE while observing the guidelines of AAMI:TIR 45.

As part of the workshop, we presented 6 possible software development life cycle (SDLC) approaches that are used for developing MMAs. Based upon the 2 previous group discussions concerning MMA development challenges and identifying MMA development framework features, it was agreed by participants to categorize each SDLC approach in terms of the following criteria: (1) achieving regulatory compliance and producing the required documentation, and (b) time to market – does the approach assist the MD to be marketed in a timely manner (see Table 4). In the following section, we outline the discussion surrounding each software development approach.

4.4.1 | Waterfall or V-model

The intention of plan-driven software development methods is to improve the product quality by reducing the number of errors that are made, while at the same time, supporting the achievement of delivery dates, budget constraints, and in the case of MD software development,

the achievement of the primary goals of creating safe and effective MDs. Medical device software developers typically develop software in accordance with a plan-driven sequential SDLC, such as the V-model or waterfall,²⁹ as it is a tried and trusted life cycle approach within the industry for producing the deliverables demanded by the regulatory authorities.^{30,31} However, participants agreed that this is a time-consuming, resource-intensive, and rigid plan-driven approach that is not well suited for developing MMAs in a timely manner.

The group discussion identified several concerns with adopting either a waterfall or V-model SDLC adoption, such as: there is no focus upon iterations, consequently, making it difficult to deliver frequent releases and to change incorrect decisions in a timely manner; they do not embrace change easily, thus, any changes introduced once the project has started can create financial overruns; they do not include a prototyping practice, therefore making it more difficult to engage users in the development process; a large amount of time is spent producing and verifying documentation, therefore leaving less time for the development and testing; there is a sign-off process after every software development phase, hence, bringing an inflexibility in terms of incorporating changes post sign-off.

In traditional MD development, there is a heavy emphasis on documentation and sign-off before proceeding to the next stage. This can lead to delays when changes need to take place. Such development life cycles are waterfall in nature with each stage having to be completed

TABLE 4 Model Comparison

Software Development Lifecycle	Regulatory Compliance and Documentation	Time To Market
Waterfall or V-model	One standard at a time—very lengthy process	Very slow; resource intensive
MDevSPICE (waterfall or V-model)	Shorter process—all standards easily put in place	Slow; resource intensive
V-model with agile practices	One standard at a time; challenge with documentation	Bit faster; full lifecycle covered
V-model with agile practices & MDevSPICE	All standards easily put in place; challenge with documentation	Faster and all standards are covered
Reduced V-model with agile practices	One standard at a time; challenge with documentation	Faster; focused on essential activities
Reduced V-model with agile practices and MDevSPICE	All standards easily put in place; challenge with documentation	Faster; focused on essential activities

prior to the next commencing. Therefore, the key concern in relation to this approach is that delivering working software may take a long time.

4.4.2 | MDevSPICE

Similarly, to MD software standards, MDevSPICE does not specify the development method that should be used in the design and development of MD software. Yet, given the fact that the following are required for MD regulatory approval: a defined process description; sign-off on all tasks and requirements; and documented evidence of these, this suggests that either a waterfall or V-model life cycle approach is well suited.

Participants felt that a significant advantage of MDevSPICE is that the software processes necessary to satisfy regulatory requirements are identified from the outset and implemented accordingly. While this approach involves the production of a lot of documentation, it streamlines the identification of appropriate software development practices. It was decided by participants that despite MDevSPICE streamlining the procedure for obtaining regulatory compliance and generating the required documentation, a plan-driven implementation is still time consuming and resource intensive. Participants felt that increased agility permits a more rapid adaptation to changing requirements and allows early detection of errors through prototyping, which fits the MMA development better than plan-driven methods.

4.4.3 | Introducing agile practices into the V-model life cycle (not based on MDevSPICE)

"Agile development focuses on achieving personal, technical, and organizational successes."³² Agile software development reports "the promise of improved software quality and reduced delivery times."³³ There are several reported benefits for adopting agile development practices including the following: more rapid feedback and adjusting and embracing change accordingly; producing frequent releases of working software; and reducing formal organizational red tape.

An agile approach for software development for MDs goes back to the 2000s, and in 2005, Sutherland evaluated Scrum, which is an agile project management approach, by using a medical company.³⁴ Scrum is currently implemented in several MD software development companies.³⁵⁻³⁸ Scrum is widely used due to its clear roadmap; its structure is easy to understand and to implement. However, Scrum is only one part of the solution, as it focuses on project management and collaboration by concentrating on business value through demonstration and prioritization. Therefore, eXtreme Programming can be used in conjunction with Scrum.² Case studies^{35,39-41} have emphasized that following an agile approach can overcome the issues associated with plan-driven software development.⁴²

Participants felt that by integrating agile practices into the V-model development life cycle, it provides the advantage of accommodating changes more easily than the previous 2 approaches. However, as this approach is not based on MDevSPICE, there is an overhead on ensuring that all the regulatory practices with the necessary documentation are in place. McHugh et al²⁶ found that by integrating agile practices into the V-model to streamline requirements changes, the amount of rework required is reduced and delivering MD software is achieved in a more timely manner.

4.4.4 | Integrating agile practices into the V-model (based upon MDevSPICE)

Alongside the benefits that an agile approach brings, there are also some challenges, such as, the need for appropriate tool support for generating documented evidence of all iterations; a proper level of traceability, risk mitigation, verification, and validation of the processes; and the need to provide detailed documentation (eg, detailed test steps⁴¹).

This approach is based upon MDevSPICE and includes agile practices that may be used to fulfil the requirements defined in the framework. Participants thought that this approach assists companies to ensure that they have all the necessary standards in place including all the documentation gathered during all stages of the life cycle. However, participants felt that there is still a challenge in relation to determining how to capture this documentation when integrating agile practices. This approach supports getting the product to the market faster than the previous approach as guidance is provided to assist organizations to put the required standards in place.

4.4.5 | Integrating agile practices into reduced V-model (not based on MDevSPICE)

This approach integrates agile practices into a reduced V-model life cycle that only includes essential activities for MMA development. The participant group found that by not integrating MDevSPICE, additional attention is required to ensure that adequate coverage of standards is in place within the implemented agile practices. It was felt that as this approach is light on documentation, extra attention should be paid in terms of what documentation is required for regulatory compliance. Participants thought that while agile practices allow working MMA software to be delivered faster, the process of implementing one standard at a time will slow the process down significantly.

4.4.6 | Integrating agile practices into reduced V-model (based upon MDevSPICE)

This approach is similar to the previous approach except it is based upon MDevSPICE. This streamlines the process overhead of ensuring that implemented agile practices satisfy all the required MD guidelines and also reduces the implementation time requirement. The authors proposed to create a mapping of the most applicable agile practices within the SDLC against the complete MDevSPICE. Then, populate the reduced V-model that is based on MDevSPICE with the most suitable agile practices to fulfil the minimum set of requirements for MMA development. This approach is light on documentation, only including the most essential documents to satisfy the regulatory requirements. Crucially, it was felt by participants that by adopting this approach, MMA software will reach the market faster than the other approaches, as developers are only focused on essential activities, with regulatory compliance being achieved seamlessly during development.

The conclusion from this discussion was that "integrating agile practices into reduced V-model" was considered to be the best MMA development approach for the workshop group participants.

The next section is going to discuss one of the key criteria identified in the discussion on features for inclusion in an MMA

development framework, which is traceability. This is an example of the action taking stage within the action research methodology

5 | TRACEABILITY

Traceability is the ability to establish and use the links (or traces) between source artefacts and target artefacts,⁴³ and it is important as developers need to ensure that their software conforms to customer expectations, functional, and regulatory requirements. Furthermore, organizations that produce safety critical software, such as those operating within the MD domain, must ensure their software is safe, reliable, and available. Traceability is an important component in ensuring safe and reliable software and is a required component of the approval and certification process,⁴⁴ which requires that safety critical software is certified before entering service.

As in other domains, traceability implementation within the MD domain is challenging and has been highlighted by a recent analysis of the traceability documents submitted to regulators in the United States as part of the MD approval process. This analysis has revealed that the traceability data was incomplete, incorrect, and conflicting in many cases.⁴⁴

Therefore, to assist MD software organizations implement traceability effectively, the authors have developed a TAIF. The TAIF consists of a traceability assessment model (TAM) and a traceability roadmap (TR). The TAM allows for the identification of any weaknesses in an organizations existing traceability process, and the TR provides the pathway for an organization to address those weaknesses. We define a TR as a set of steps or activities that, if followed, will provide guidance to an organization implementing traceability and ensure that it is compliant with the traceability requirements from the MD standards and traceability implementation best practices.

An overview of the approach used to develop the TAM and its structure is presented in Section 5.1. Further detail on the structure of the TAM can be obtained in the previous studies.^{45,46} Section 5.2 presents an overview of the approach used to develop the TR along with its structure. Further detail on the structure of the roadmap can be obtained in Regan et al.⁴⁶ Section 5.3 provides an outline of the approach used to validate the TAIF in addition to a summary of the findings from implementing the TAIF within 2 MD software organizations.

5.1 | Traceability assessment model development and structure

It was decided to base the development of the TAM on the ISO/IEC 15504 (SPICE) assessment model mainly because ISO/IEC 15504 is used extensively in other safety critical industries such as the automotive industry (Automotive SPICE), space industry (SPICE 4 SPACE), and the MD industry (MDevSPICE).

Having decided to base the TAM on IS/IEC 15504, an approach to its development was formulated as depicted in Figure 2.

A process assessment model (PAM), as defined in ISO/IEC 15504, is based on a 2-dimensional model containing a process dimension and a capability dimension. The process dimension is provided by a PRM, which defines a set of processes characterized by statements of process purpose and process outcomes.

The development of the PRM began with an analysis of the MD standards to extract the traceability requirements within those standards. The standards analyzed were ISO 13485; IEC 62304; ISO 14971; IEC 80002-1; IEC 60601-1; and the FDA guidance documents on MD software development, which includes the guidance document on the development of MMAs. This analysis resulted in 24 traceability requirements. Additionally, as the aim of this study is to develop a TAIF, which will assist with the implementation of the standards' requirements for traceability along with the best practices for implementing traceability, the 23 best practices identified during a literature review were included in the PRM.⁴⁷

So as to establish a uniformity of description of the processes in this TAM, technical report ISO/IEC TR 24774 (ISO/IEC, 2010) used. This report provides guidelines for process description.

Section 6.2 of ISO/IEC 15504-2 sets out the minimum requirements for a PRM. These requirements state that the process descriptions incorporate a statement of the purpose of the process together with the set of outcomes, which demonstrate successful achievement of the process purpose.

While ISO/IEC 15504-2 details the minimum requirements that a PRM and a PAM should meet, it provides no guidance on how to develop the models, ie, it does not tell you how to transform a set of requirements into a PRM or PAM. So as to address this, issue the Tudor IT Service Management Process Assessment (TIPA) transformation process⁴⁸ was used as it complies with the requirements for PRMs and PAMs as expressed in ISO/IEC 15504-2. The TIPA process takes the requirements produced by the medical device standards and

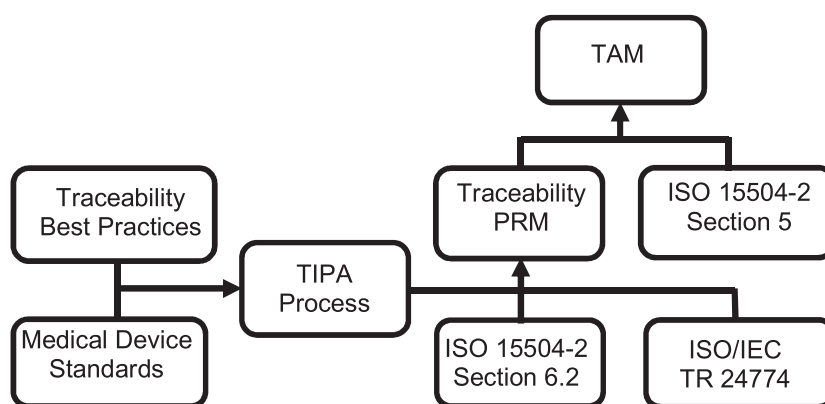


FIGURE 2 Traceability assessment model (TAM) development approach. PRM, process reference model; TIPA, Tudor IT Service Management Process Assessment

traceability best practices and transforms them into milestones and goals.

The TAM contains the following 4 traceability processes: SDLC traceability process; risk management traceability process; change management traceability process; and a traceability implementation best practices process. In accordance with ISO 15504, each of the 4 processes in this TAM contain a process description and a set of indicators, known as base practices and work products. Base practices provide a definition of the tasks and activities needed to accomplish the process purpose and fulfil the process outcomes. Evidence of performance of the base practices and the presence of work products provide objective evidence of the achievement of the purpose of the process.

5.2 | Roadmap development and format

The approach used to develop the TR is depicted in the figure.

The roadmap development methodology depicted in Figure 3 provides a systematic approach to roadmap construction.⁴⁹ The first 3 steps of this method involved identifying traceability requirements from the standards, grouping the requirements into goals, and then separating the goals in line with ISO/IEC 15504 capability levels. To develop the roadmap, it was not necessary to complete these steps as they had already been completed as part of the development of the TAM. The grouped goals or milestones identified in the above process are equivalent to the grouped goals contained within the TAM and so a milestone equates to a TAM process.

Step 4 involved ordering the milestones based on the capability level and logical groups. The 4 processes within the TAM are all at capability level 1 and so conferred no order priority. Thus, the processes were arranged in the order in which they should be implemented within a development project. This order consists of the best practices being implemented before development begins. Then the RM trace process and CM trace process are to be implemented concurrently with the SDLC trace process.

Steps 5 and 6 consisted of validating the generated roadmap (by industry experts) and identifying activities that can meet the identified goals. It was considered to be more productive to alternate these 2 steps, as the TAM processes had already been validated by expert review during the development of the TAM. Therefore, activities were identified before the roadmap was referred for initial validation by expert review. These activities are labelled as base practices in the roadmap, with each base practice containing 2 exemplar methods of how to implement them. For the final step, the TR underwent industry

validation by implementation in 2 medicals device software organizations.

A literature review on roadmaps revealed a taxonomy of 8 roadmap types for purpose, and 8 roadmap formats for graphical representation.⁵⁰ It was considered that the TR would better align with the “process planning” type of roadmap. This type supports the management of knowledge, focusing on a particular process area. In the case of the TR, the focus is on knowledge flows that are needed to facilitate effective traceability implementation. In addition to, and perhaps more important than roadmap type, the format of the roadmap was then considered. It was decided that the roadmap should be a hybrid version entailing mainly “single layer” and “text” formats, with some pictorial representation. The pictorial representation is presented early in the roadmap to give the user an overview whereas text is used to describe the roadmap in detail. This hybrid version was chosen for the following reasons:

- Combinations of pictorial and text information facilitate significant improvements in understanding⁵¹;
- The single layer roadmap is a subset of the multilayer format and is less complex. A pictorial representation can present a lot of information in a readily understandable format.

5.3 | Traceability assessment and implementation framework validation

The evaluation stage of the action research methodology involved an evaluation of the TAIF. The approach taken to its' validation is presented in Figure 4. The validation is completed in 2 stages. The first stage is an initial validation by expert review while the second stage is further validation by implementation of the TAIF in 2 MD organizations.

5.3.1 | Initial validation

An initial validation of the TAIF was completed through expert review.^{45,46} Thirteen international experts in areas such as ISO/IEC 15504, MD software development and processes, MD software standards, traceability research and implementation, software process improvement, and roadmap development. This validation entailed the experts being asked to review the models and then to complete a questionnaire about the models. The questionnaire was composed of questions relating to the usefulness, usability, and content of the models and asked for suggestions for improvement. A focus group was convened, comprising of 3 members with great experience in

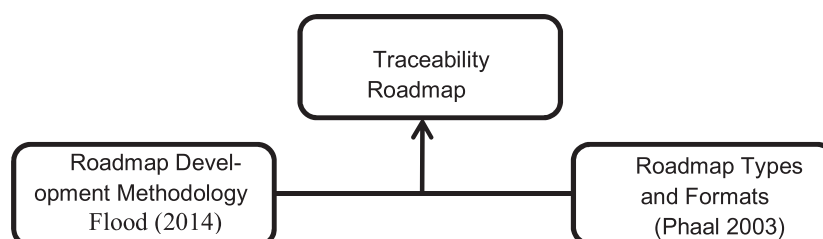


FIGURE 3 Roadmap development approach

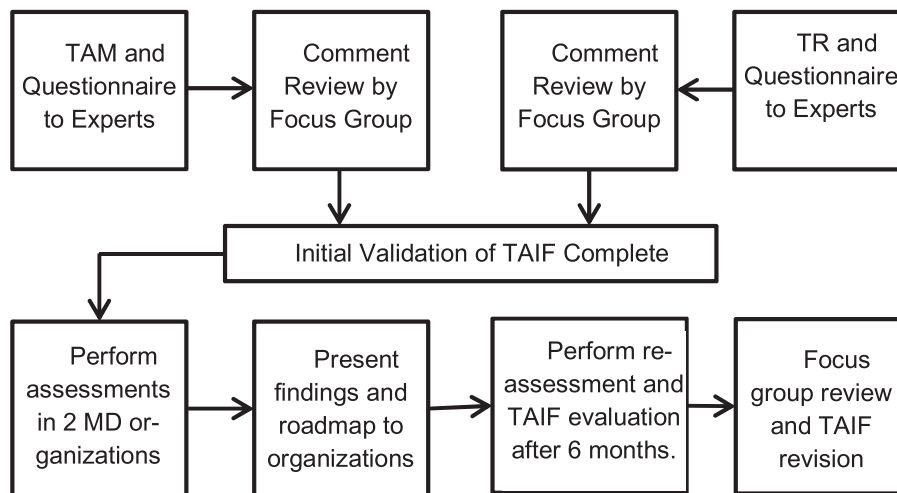


FIGURE 4 Traceability assessment and implementation framework (TAIF) validation approach

developing questionnaires, to validate that the questionnaire would achieve its aims. After the questionnaires were returned, another focus group was formed to consider all the responses and through general consensus decisions were taken as to what changes should be made to the models. After the changes were made, the models were now ready for industry trial.

5.3.2 | Approach to industry validation

For us to further validate the TAIF, a longitudinal study of 2 MD organizations was performed over a 6-month period. An initial assessment was conducted to assess the organizations' implementation of traceability requirements and best practices. One week later a report was presented to both organizations detailing their traceability weaknesses, along with a roadmap, which provided recommendations on how they could address those weaknesses. After a 6-month period a follow-up assessment was conducted. This follow-up assessment ascertained if any improvement had been made. Additionally, the follow-up assessment ascertained each organization's perspective of the assessment experience and of the roadmap.

For us to collect the relevant data a mixture of interviews and document reviews were used. Interview questions to ascertain the current implementation of traceability requirements and best practices were developed based on the TAM base practices. So as to determine the organization's perspectives of the assessment experience and the roadmap, questions were formulated which assessed their value, usefulness, and usability to the organizations. A semistructured interview was developed based on the interview guide development strategy championed by Dawson.⁵² The interview contained a mixture of both open and closed questions. The closed questions ensured that very specific and required data was collected, eg, the organizations implementation of specific traceability requirements. The open questions allowed the interviewees to reflect and give their opinion on the value, usefulness, and usability of the assessment experience and the roadmap. These questions were then independently reviewed by 2 persons with extensive experience in developing interview guides.

So as to introduce and implement the TAIF to both organizations, the Adept method⁵³ was used. The Adept method was used as it could

provide a comprehensive assessment of an organizations' traceability compliance without being too onerous in terms of resources that the organization needs to provide. In addition, the Adept method also includes a follow-up study to evaluate how well the implementation of the recommended changes was performed allowing for a full evaluation of not only the assessment model but also the roadmap.

5.3.3 | Sample size and participants

This study was conducted in 2 MD organizations. The main factors that directly influenced the sample size for this study were the following:

1. Resources in terms of time and cost.
2. Perceived requirement. The authors considered this study has a narrow topic and uses in depth interviews which will limit the number of participants needed. Additionally, the general population of MD software SMEs in Ireland is approximately 50, which again limits the number of participating organizations required.

The 2 participating organizations were chosen on the basis that the researcher was able to get personal introductions to management within these organizations. Two individuals from each organization participated in the interviews. These individuals best understood each organizations traceability process and were selected by the organization themselves.

The first organization that agreed to participate in the study, Organization A, is a small MD software company, founded in 2002, based in Ireland. The company has a total of 10 employees, which include 1 programmer, 1 software tester, and 1 quality assurance person. The company produces MD software with a software safety classification of B, meaning nonserious injury is possible.

The second company, Organization B, is a small product development and design engineering company focused on the MD and Life Science market and is based in Ireland. The company, which was formed in 2007, employs 14 individuals with skills in mechanical, hardware, and software engineering. The company are a third party supplier of software to MD companies and have recently been accredited with IEC 62304 certification.

5.3.4 | Data analysis

The interviews, which were recorded by Dictaphone, were transcribed into Microsoft word documents. Additionally, notes were made during the interview. Because of the the open-ended nature of part of the interview guide, the data generated consisted of long paragraphs with similar concepts in different locations of the text and with data unrelated to the study. This data was considered to be unstructured, and so qualitative content analysis, using deductive and inductive coding, was used to structure the data into meaningful and analyzable data from which conclusions could be drawn. Categories and codes based on the purpose of the interview questions were established prior to coding (deductive). Annotating the text involved carefully reading the text a number of times, each time underlining the text that applied to any category and annotating the underlined text with the category code. While conducting this process, a number of new categories emerged (inductive). This process was repeated until no new text for coding emerged.

5.3.5 | Implementation findings

The “specify learning stage” of the action research methodology involves the presentation of the validation findings. A comparison of the initial and follow-up assessments indicates that both organizations had significantly improved almost all traceability processes, although none of the processes improved to a level of 100% adherence. For further details on the findings and how the traceability improvements were measured, refer to Regan et al.⁵⁴ The overriding reason given for this is resource issues (mostly time) within the organizations. These resource issues are clearly outside the scope of the TAIF.

An evaluation of the assessment experience and TR clearly indicates that both organizations found them to be very useful, usable, and generally providing an awareness, knowledge, and expertise in traceability that they did not previously have. Additionally, both organizations considered that the assessment provided them with guidance and real direction for long-term traceability improvement.

6 | CONCLUSION AND FUTURE WORK

This paper described our approach for an MMA development framework. To ensure that the framework addressed the true needs of the MMA development community, we conducted a workshop with the experts within the MD software domain. Within this workshop, there was 3 discussion areas. The first of these areas concerned challenges within MMA development. The second area of discussion concerned the features that should be included within an MMA framework to overcome the challenges mentioned in first discussion. The final discussion concerned the actual software development approach for the MMA framework where we looked at 6 different software development life cycles.

While the MMA framework will contain many features that were identified within the workshop by experts, due to space restrictions, we have provided an exemplar of one of these features—traceability. We described the development and validation of the traceability component of the MMA framework.

In the future, we plan to develop the complete MMA framework to include all of the features identified in Section 4.3. Upon completion of this MMA framework, we plan to validate it within the organizations that participated in the workshop. Another area we wish to focus upon is the development of an agile approach for MDevSPICE that can be used for both MMAs and general MD software development. This work will be validated both within the agile and MD standards communities and the MD software industry.

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